# Table of Contents

## 1. Introduction

This is Introduction Lab, a lab meant to reintroduce various concepts to the student such as the superposition principle. This lab took place on January 22, 2025. The signed pages are verified by the TA at the end of the report in the Appendix.

## 2. Objectives

The objective of this lab was to determine the combined RMS output voltage value of the circuit using the formulas used in the prelab as well as the input voltages as AC and DC voltages as well as the output voltages to calculate the total RMS values.

## 3. Circuit Under Test

**Figure 1** displays the entire complete circuit's design that is to be tested. The left side of the circuit consists of a few resistors connected in series with a capacitor supplied with a signal generator. The right side of the circuit consists of two resistors connected in parallel with a voltage $V_{CC}$. This can be seen in **Figure 2a** and **Figure 2b**. This figure displays the circuit when the capacitor acts as an open circuit. **Figure 3** displays the circuit when the left side of the circuit and the right side of the circuit are connected and resembles when the capacitor is shorted. Altogether, similar to **Figure 1**, **Figure 4** is the left and right side of the circuits connected with the 100nF capacitor.

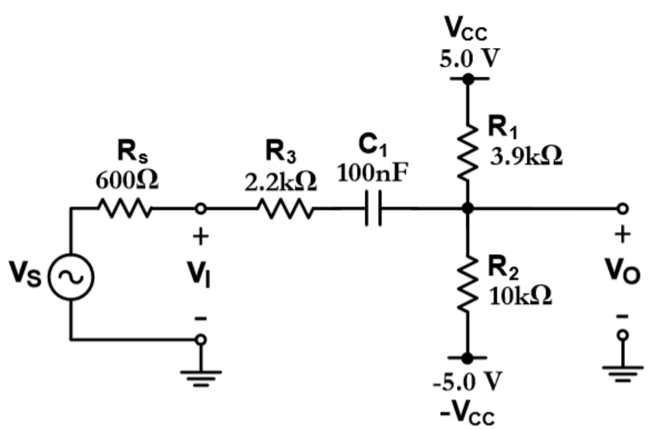


**Figure 1. A circuit involving both AC and DC voltages.**

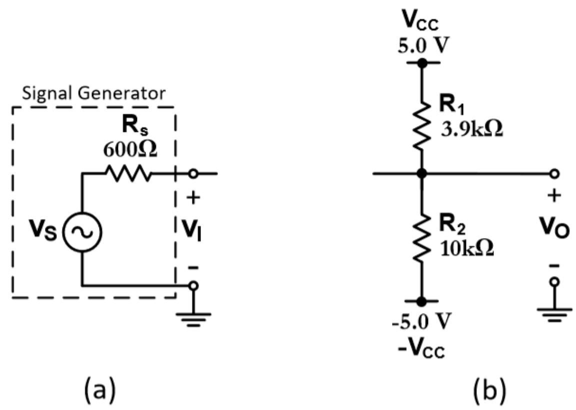


**Figure 2. Circuit for Step E1.**

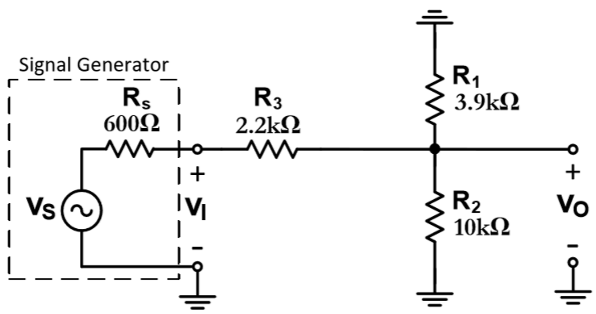


**Figure 3:** Complete circuit based on **Figure 1** with shorted capacitor.

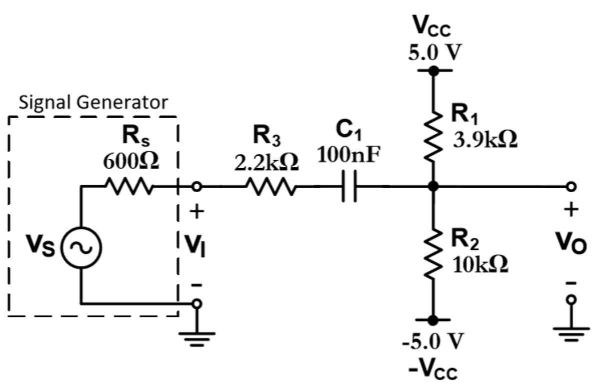


**Figure 4:** Complete circuit based on **Figure 1** with capacitor.

## 4. Experimental Results

The circuit displayed in **Figure 1** was constructed on a breadboard and was simulated with the use of a signal generator, oscilloscope, and power supply. The left subcircuit and right subcircuit were measured separately initially to get $v_I$ and $v_O$ values. The left subcircuit was simulated according to **Figure 2a** using a multimeter to get values in **Table E1** with a supplied frequency from the signal generator at 20 kHz with a peak to peak voltage of 4 volts and 50 ohm resistance supplied. Connecting the signal generator to a 560 ohm resistor, a multimeter measured the voltage across the resistor to get the $v_I$ voltages in **Table E1**. The right subcircuit was then set up according to **Figure 2b** by supplying 5 volts and -5 volts to the resistors respectively. $v_O$ values from **Table E1** were measured by connecting the multimeter across the common node and ground. **Table E2** was simulated similarly according to **Figure 3** by completing the entire circuit and measuring the voltage values in the table and the generated waveform for this circuit can be seen in **Figure 5**. **Table E3** was simulated similarly according to **Figure 4** with a capacitor with the same measuring method using a multimeter. The generated waveform for this circuit can be seen in **Figure 6**.

**Table E1:** Table of results corresponding to **Figure 2a** and **Figure 2b** as an open circuit at the capacitor**.**

| $v_s$ (p-p) | $v_I$ (rms) | $v_I$ (dc) | $v_O$ (rms) | $v_O$ (dc) | $v_O$ (RMS) |
|---|---|---|---|---|---|
| 4 V | 2.847 V | 0.001 mV | 0.621 V | 2.95 V | 3.01 V |

**Table E2:** Table of results corresponding to **Figure 3** as a short circuit at the capacitor**.**

| $v_s$ (p-p) | $v_I$ (rms) | $v_I$ (dc) | $v_O$ (rms) | $v_O$ (dc) | $v_O$ (RMS) |
|---|---|---|---|---|---|
| 4 V | 2.836 V | 0.153 mV | 2.827 V | 2.18 V | 3.56 V |

**Table E3:** Table of results corresponding to **Figure 4.**

| $v_s$ (p-p) | $v_I$ (rms) | $v_I$ (dc) | $v_O$ (rms) | $v_O$ (dc) | $v_O$ (RMS) |
|---|---|---|---|---|---|
| 4 V | 2.810 V | 0.001 mV | 2.810 V | 2.192 V | 3.56 V |

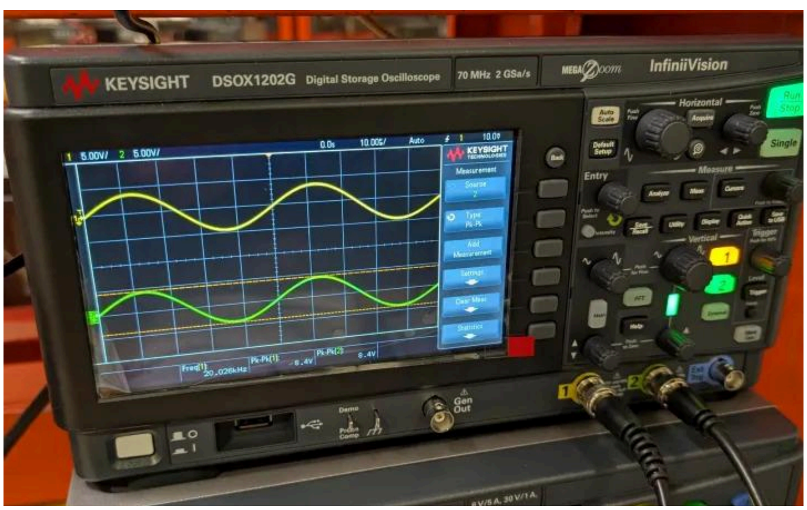


**Figure 5:** Simulated waveform of circuit from **Figure 3**, Graph E2.

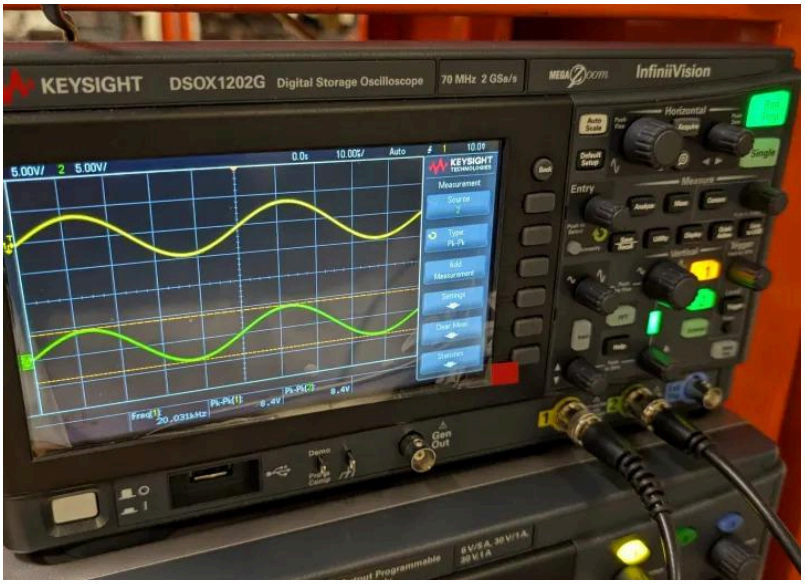


**Figure 6:** Simulated waveform of circuit from **Figure 4**, Graph E3.

## 5. Conclusions and Remarks

**C1. Compare the results of Table P1 and Table E3, and comment on the reasons for discrepancies, if any. Do the results agree with the equation presented in Part P1 for the rms value of a composite signal?**

Comparing **Table P1** and **Table E3**, it can be seen that the output rms and dc voltages match, however there is a discrepancy between the input rms voltage across the tables. This may be due to fluctuations in the power supply or a difference in the resistance of the 560 ohm resistor. Due to the 50 ohm resistance present in the signal generator, a 560 ohm resistor was set to be in series with the signal and the 10 ohms of resistance difference was neglected. Overall, the results agree with the rms value of the composite signal and the equation presented in Part P1. The values between the two tables align with each other and have a percent error of 11.37%.

**C2. Justify the waveform of *vO* in E3, based on the principle of "superposition" and what you observed in experiments E1 and E2.**

The waveform of $V_O$ in E3 can be justified with use of the superposition principle. The superposition principle states that the response in a linear circuit with various independent sources is the sum of each of the responses acting alone. This can be seen in the case of the peak to peak voltage as it is around the sum of the $V_O$ RMS values from experiments E1 and E2. If they are to be compared, the sum of the RMS $V_O$ values and the peak-to-peak voltage have a percent error of 21.7%. This signifies the relationship of the independent sub circuits and proves the superposition principle.

**C3. Using Multisim or any other circuit simulation software, simulate the circuit shown in Figure 1 (which is the same as the circuit in Figure 5), based on the assumption that the voltage *vS* is a 20-kHz sinusoidal signal with a peak-to-peak swing of 4 V. In the lab report, attach the resulting simulated waveforms for *vS*, *vI*, and *vO*, all plotted on one single frame and labeled clearly. Additionally, include the schematic diagram of the simulated circuit by printing from the screen in Multisim. Compare the waveforms obtained from the simulation with those of Graph E3, and comment.**

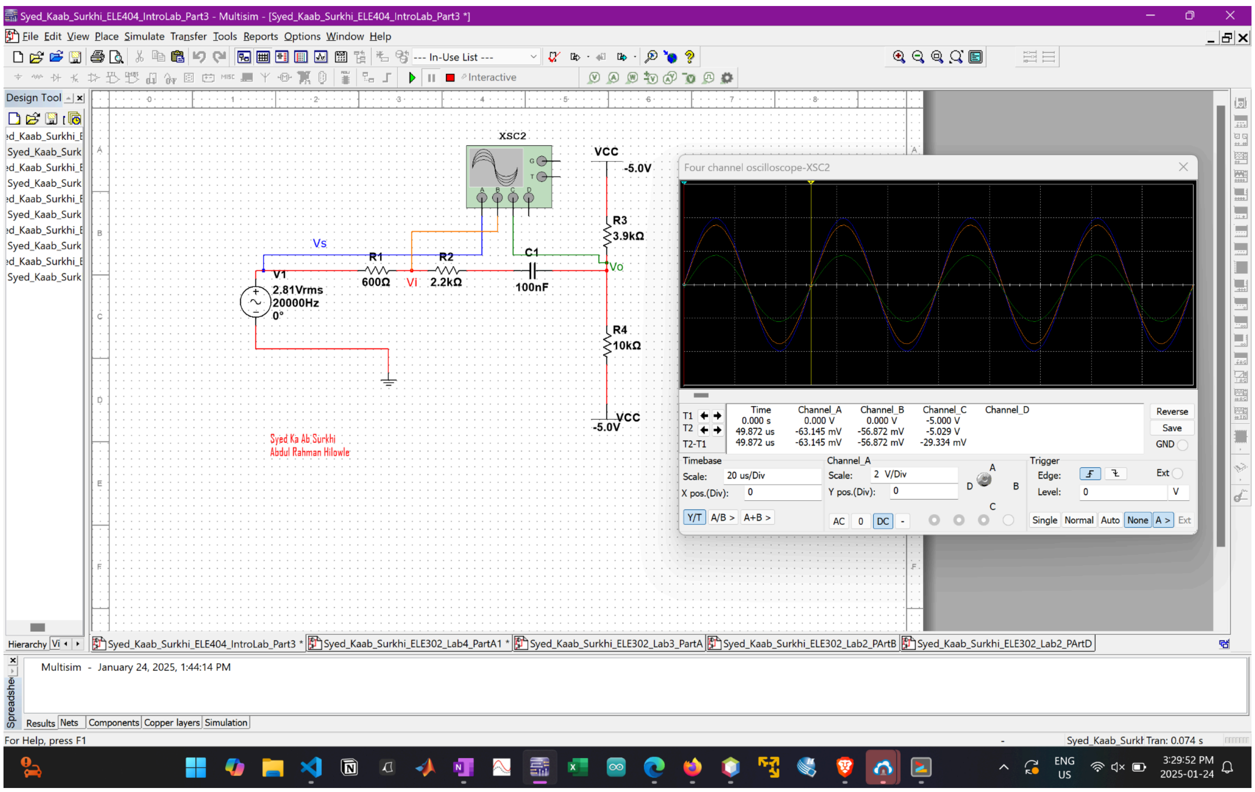


**Figure 7:** Simulated waveform on MultiSim based on **Figure 1** where Channel_A is $V_S$, Channel_B is $V_I$ and Channel_C is $V_O$.

Comparing **Figure 7** with Graph E3 in **Figure 6**, it can be seen that the $V_O$ waveforms somewhat align however the $V_I$ waveforms do not align. This may be due to the same discrepancies mentioned earlier in C1. There may have been a discrepancy in the resistance of the provided resistor and the negligence of the excess resistance may have contributed to the difference in the amplitudes of the waveforms of $V_I$.

# 6. Appendix